Lecture plan · for the lecturer

Causal inference: DAGs and difference-in-differences

Data science · University · Lecture 3

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Spot confounding.
- Choose an adjustment set from a DAG.
- Estimate an effect with DiD.

Activity: Draw it, then estimate it

1. Warm-up (5 min): هل نغلقه؟
2. Teach (15 min): Draw the tutoring DAG and mark confounders, mediators and colliders.
3. Draw (15 min): Teams draw a DAG for a headline claim and choose an adjustment set.
4. Estimate (20 min): Compute a DiD from a two-by-two table, then plot pre-trends and discuss whether they convince.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- What would make you trust the tutoring estimate?
- Why is a collider dangerous to adjust for?
- How will you grade assumptions?

Exit ticket

- Draw a three-node DAG and say what you would adjust for.

Try it in your next lecture
Give teams one news headline that claims a cause; ask them to draw the causal diagram and name one confounder the article ignored.

Seminar sheet · for students

Causal inference: DAGs and difference-in-differences

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Name _____ Class _____ Date _____

1 **Treated: 50 to 58. Comparison: 52 to 55. What is the DiD estimate?**

◯ 8 points ◯ 5 points ◯ 3 points

2 **Work out the difference-in-differences.**

Treated: 40 → 49. Comparison: 45 → 48. The treated change is _____, the comparison change is _____, so the effect is _____.

Word bank: 9 · 4 · 12 · 3 · 6

3 **To estimate the effect of tutoring on grades, which should you adjust for? Choose all that apply.**

☐ Prior grade, which affects both tutoring and grades ☐ Hours studied, which tutoring changes

☐ Year of study, which affects both ☐ Being on the dean's list, caused by both tutoring and grades

4 **Parallel trends can be confirmed by checking that the two groups had similar trends before the treatment.**

◯ True ◯ False

5 **Match each design to the threat it most directly addresses.**

Randomised experiment	• • • •	Hidden, unstated assumptions
Adjusting for measured confounders	• • • •	Fixed group differences and common shocks
Difference-in-differences	• • • •	Back-door paths through observed variables
Drawing the DAG first	• • • •	All confounding, measured or not

✎ **Which causal claim do your students meet most often in your field? What diagram would you ask them to draw for it?**